# 10/14/2026: Induction

CSCI 246: Discrete Structures

Textbook reference: Scheinerman Sec. 22

## Midterm Pickup

Midterms are in the front of the room, grouped into four bins (A-G, H-L, M-R, S-Z) by last name. The midterms are upside down with your last name on the back. Come find yours before, during, or after class. Only turn the quiz over if it's yours.

## Motivation: This course and the current job market

- **Mathematical reasoning is harder for AI to replace.** The work most exposed to automation is routine junior coding. Understanding why a model or algorithm behaves the way it does, judging whether an experiment is sound and reasoning about uncertainty are harder to automate.
- **Specialized roles are doing better than generalist ones**: For example, AI/ML engineering is up 39% since 2022.

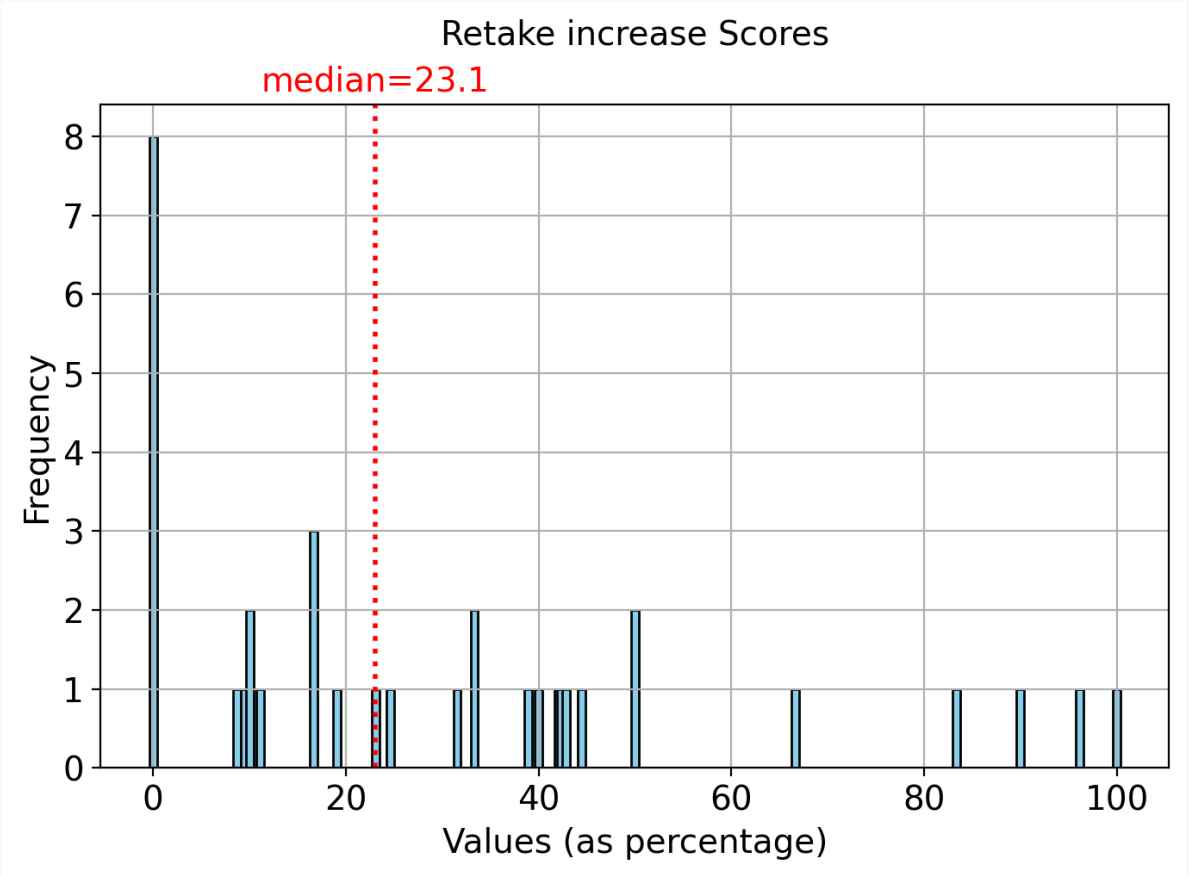
## Today's Agenda

- Quiz retake, midterm Review ($\approx$ 5 mins)
- Mini-lecture ($\approx$ 25 mins)
- Group exercises ($\approx$ 20 mins)

# Notes on Quiz Retake, Midterm

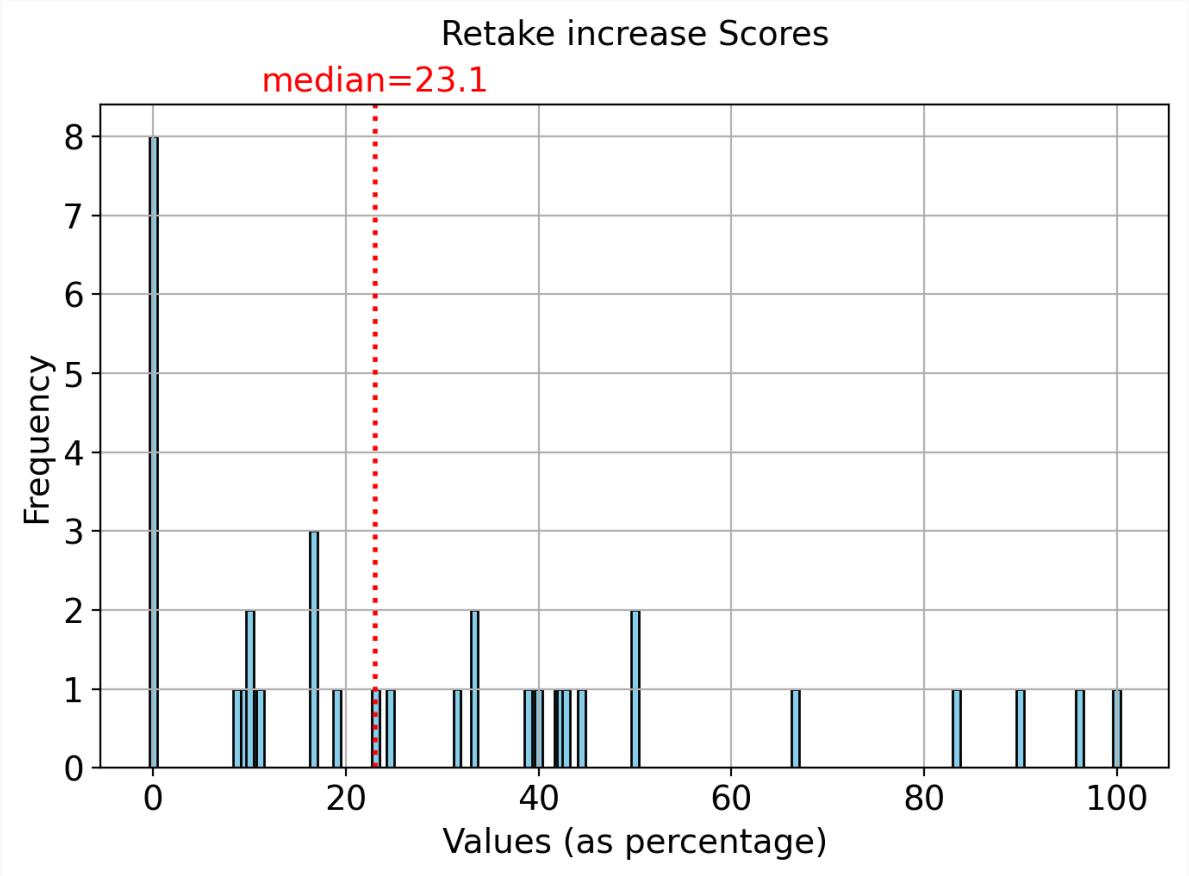
# Quiz Retake Results

You improved your scores by quite a bit!



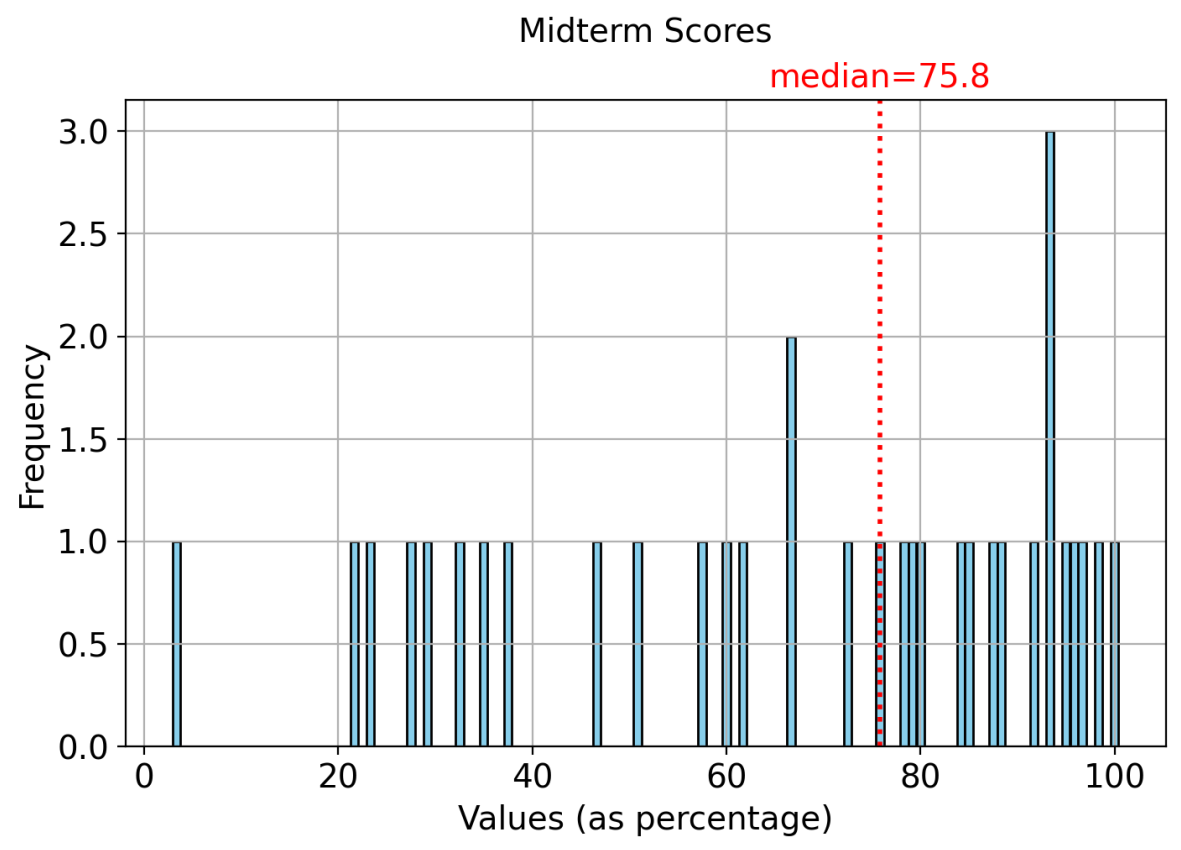
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(Apologies to those who had already maxed out their score.)

# Midterm Results



# What you could do if your midterm score is lower than you'd like?

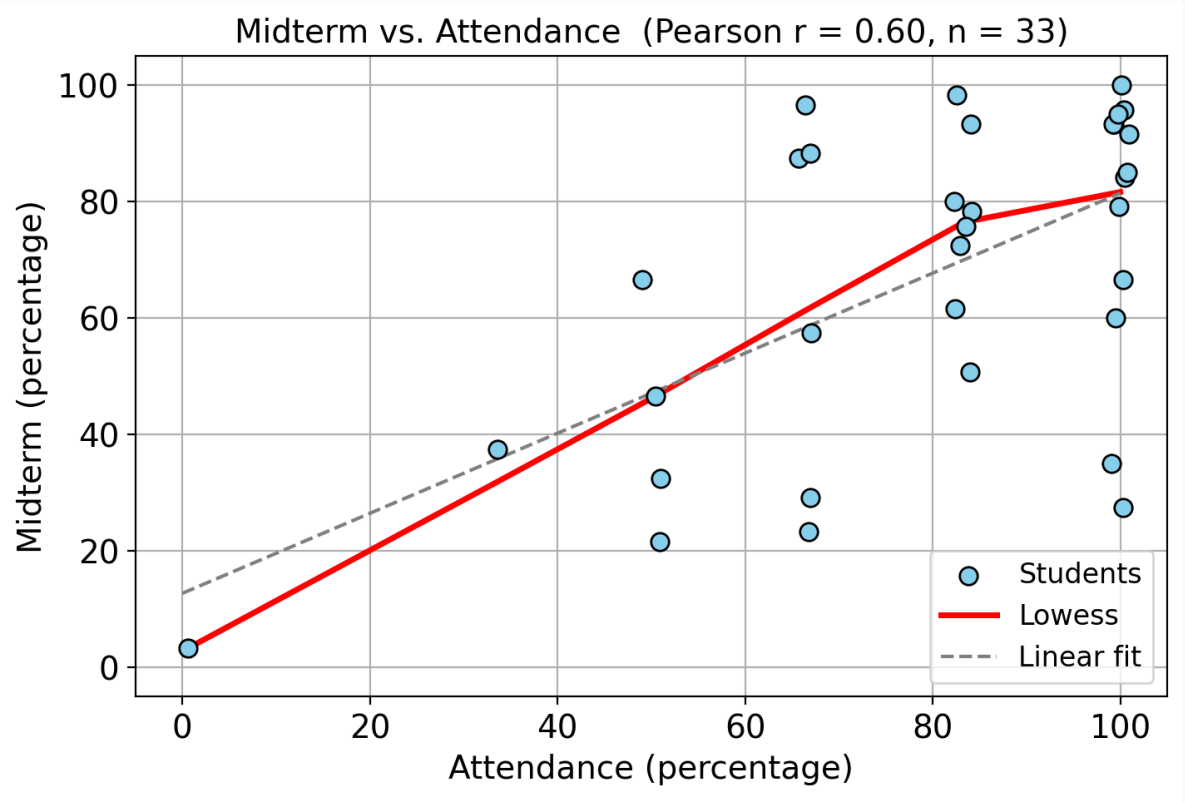
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5. Come chat with me to brainstorm on the details.

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**Opportunity.** I will offer optional midterm retakes. Details forthcoming.

# Study Guide for Quiz 6

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**Group exercises.** Know how to solve the group exercises from the course meetings on Binomial Coefficients (9/26) and Contradiction and Contraposition (10/3).

**Readings.**

Sec. 17 (Binomial Coefficients )

- Understand Def 17.1, as used in Examples 17.2-17.4 and 17.6.
- Understand how to apply the computational formula of Theorem 17.12.
- Understand the intuition behind Prop 17.7.

Sec. 20 (Contradiction and Contraposition)

- Understand how to apply Proof Template 11.
- Understand how to apply Proof Template 12, as used to prove Proposition 20.3.

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Sec. 20 (Contradiction and Contraposition)

- Understand how to apply Proof Template 11.
- Understand how to apply Proof Template 12, as used to prove Proposition 20.3.

**Extra credit proof review question.** I will give 10% extra credit on your midterm if you can answer the review question perfectly.

*Let $x$ be an integer. Prove that $0 \mid x$ if and only if $x = 0$.*

# Induction

# Induction Demo

**Good news!** I'm going to pay you to learn induction!



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**Rule.**

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| 1 | Yes | Base |
| 2 | Yes | Step applied to student 1 |

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| ⋮ | ⋮ | ⋮ |

# Triangular Numbers

**Poll**

Last class, we proved

$$1 + 2 + 3 + \cdots + n = n(n+1)/2.$$

What is the intuition behind this formula?

# Triangular Numbers

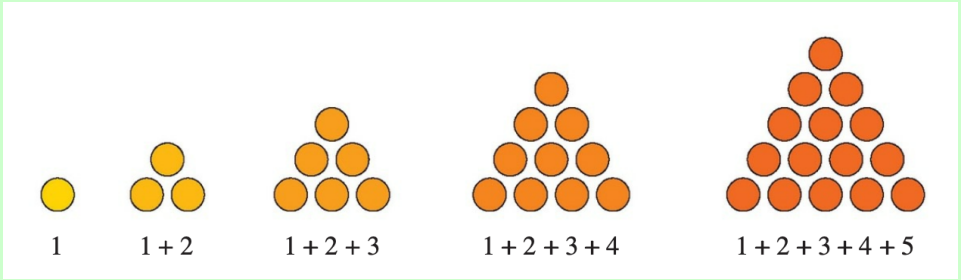
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## Hint



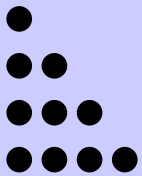
# Triangular Numbers: Geometric Intuition

## Triangular Numbers

The $n$-th triangular number is

$$T_n = 1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}.$$

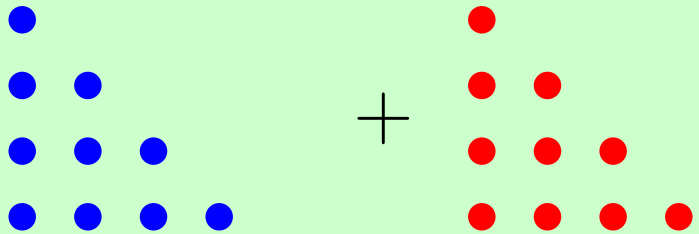
It counts the number of dots you can arrange in an **triangle** like this:



# Triangular Numbers: Geometric Intuition

## Geometric Intuition: Make a rectangle from two triangles

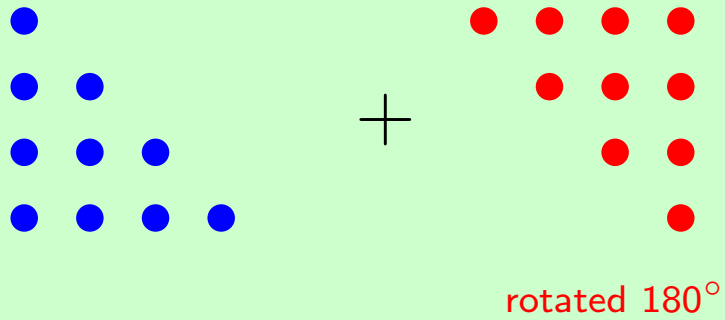
Take **two copies** of the triangle for $T_4$.



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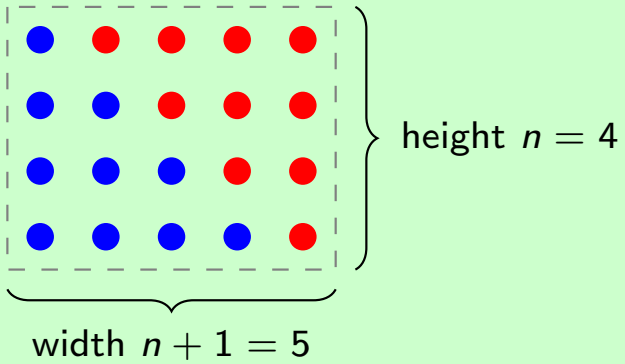
Take **two copies** of the triangle for $T_4$. Rotate one copy by $180^\circ$.



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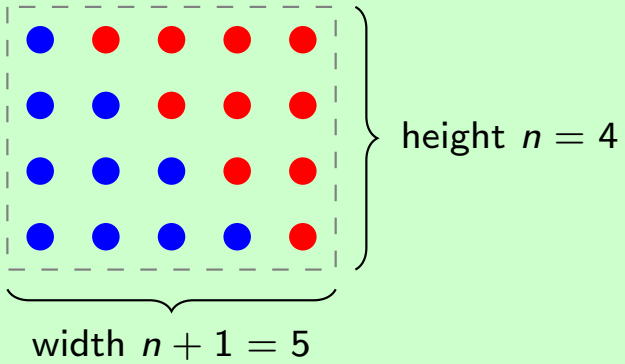
Take **two copies** of the triangle for $T_4$. Rotate one copy by $180^\circ$. Then slide it into place: the two triangles fit together into a **rectangle**.



# Triangular Numbers: Geometric Intuition

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So the rectangle has **area** $n(n+1)$ dots. But it is made of **two identical triangles**, so the number of dots in each triangle is

$$T_n = \frac{n(n+1)}{2}.$$

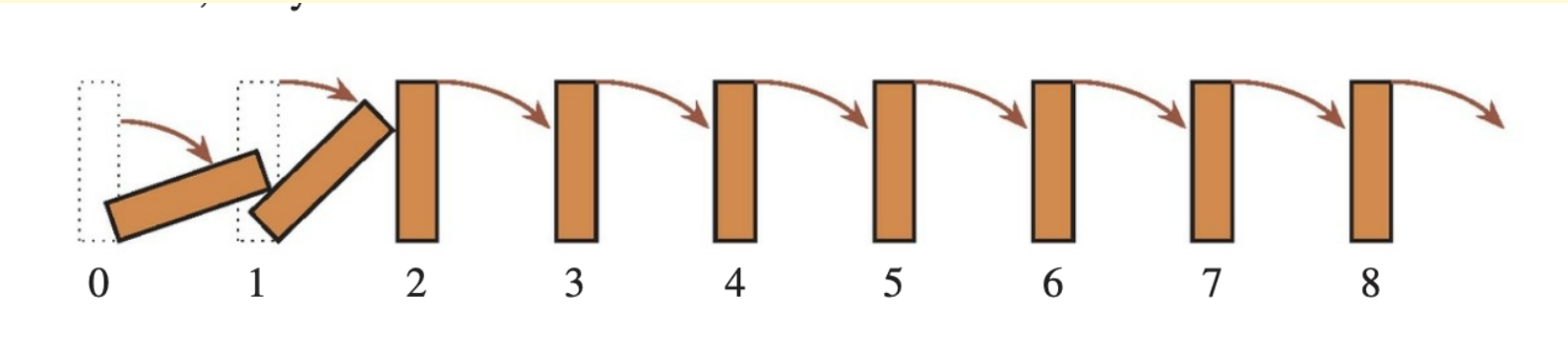
# The Induction Strategy to Proving Things

**Problem**

Prove that the $n$-th triangular number is $n(n+1)/2$. That is, prove

$$1 + 2 + 3 + \cdots + n = n(n+1)/2.$$

**Induction Strategy**




**Base.** Show the equation holds for some starting point (usually 0 or 1).
**Induction step.** Show that if the equation holds for natural number $n = k$, then it must hold for natural number $n = k + 1$.

The strategy in the yellow box lets us conclude the equation holds for all $n$.

# Proof By Induction: Example

**Problem (Induction)**

Prove that the $n$-th triangular number is $n(n+1)/2$. That is, prove

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}. \tag{1}$$

**Base.** We show Eq. (1) holds for $n = 1$. In that case, Eq. (1) says

$$1 = \frac{1(1+1)}{2}$$

which is true.

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We have

$$\text{LHS of Eq. (2)} = \frac{k(k+1)}{2} + (k+1) \qquad \text{(by induction hypothesis)}$$

$$= \text{RHS of Eq. (2)} \qquad \text{(after some algebra)}$$

# Induction and Strong Induction

What we've been using so far is classical induction.

**Scheinerman Theorem 22.2: Induction**

Let $A$ be a set of natural numbers. If

- $0 \in A$, and
- $\forall k \in \mathbb{N}, k \in A \implies k+1 \in A$

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There is also a strong version.

**Scheinerman Theorem 22.9: Strong Induction**

Let $A$ be a set of natural numbers. If

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**Remark.** If the result you wish to prove doesn't cover all natural numbers – say it covers just the positive integers – the basis step may begin at a value other than 0.

# Strong Induction

## When to Use Strong Induction

Sometimes there are scenarios where the induction implication goes naturally not from $n$ to $n+1$, but:

- From some smaller number to $n+1$, or
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## Example: Every positive integer has a prime factorization

- Suppose every number smaller than $n$ has a prime factorization.
- The number $n$ must either be prime (in which case we are done) or composite.
- If $n$ is composite, then $n = ab$ for smaller number $a, b < n$. But $a, b$ have their own factorizations by the induction hypothesis. These can be put together to make a prime factorization of $n$.

# Uses of induction in computer science: Examples

1. **Proving correctness of recursive algorithms:** E.g. we can use induction to show `factorial`$(n) = n!$.

```
1  def factorial(n):
2      """Compute n! recursively"""
3      if n == 0:
4          return 1
5      else:
6          return n * factorial(n - 1)
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8  print(factorial(5))
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3. **Proving properties of data structures:** Many structures (like linked lists, binary trees, heaps, etc.) are defined inductively.
4. **Proving correctness of loops**: You show that some property holds before the loop. Then show if it holds at one iteration, it still holds at the next.

# Group exercises

# Random group assignments

Ashley, Isabelle: 3
Bailey, Tanna: 13
Blaisdell, Olyver: 13
Brown, Ashton: 5
Brown, Jonathan: 1
Cardenas, Dominic: 11
Cutler, George: 9
Fabik, Roland: 3
Fellows, Audrey: 8
Fitzgerald, Ryan: 5
Fuller, Ryan: 6
Griffen, Clara: 10
Grutsch, Tanner: 9
Hager, Nathan: 7
Horner, Jayden: 12
Johnson, Jaret: 2
Kintzel, Samantha: 4
Lameres, Alexis: 4
Lee, Holden: 1
Mayala, Reece: 5
McLean, Connor: 7
Mcbride, Daniel: 3
Miller, Maverick: 10
Moss, Emily: 6
Mousseau, Matthew: 12
Neuman, Evan: 7
Putnam, John: 2
Ramos, Vydel: 8
Rathbun, Jedidiah: 11
Roberts, Matthew: 10
Roller, Nathan: 1
Ryan, Otis: 11
Shevtsov, Timothy: 4
Sinclair, Aidan: 8
Stensland, Emma: 2
Thiessen, Logan: 9
Thompson, Jaiden: 6
Vath, Ethan: 12

# Group exercises: Induction

1. Show by induction that $n^2 - n$ is even for any natural number $n$ (that is, for $n = 0, 1, 2, \ldots$).
2. Show by induction that $2^n < n!$ for all $n \geq 4$.
3. Show by induction that $f_0 + \cdots + f_n = f_{n+2} - 1$ in the Fibonacci sequence.
4. Show by induction that $f_n < 2^n$ in the Fibonacci sequence.

**Definition**

The **Fibonacci sequence** is the sequence given by the following recursive rules:

$$f_0 = 0, \qquad f_1 = 1, \qquad f_{n+2} = f_n + f_{n+1}$$

The sequence is: $0, 1, 1, 2, 3, 5, 8, 13, 21, 34, \ldots$

# Solutions to group exercises

## Exercise 1

Show by induction that $n^2 - n$ is even for any natural number $n$ (that is, for $n = 0, 1, 2, \ldots$).

## Solution (by Induction)

**Induction base.** Take $n = 0$. Then $n^2 - n = 0^2 - 0 = 0$, which is even. So $n^2 - n$ is even when $n = 0$.

**Induction step.** Now we assume the proposition holds at index $k$ (this is called the induction hypothesis), and we need to show it holds at index $k + 1$. That is, we assume $k^2 - k$ is even, and we need to show that $(k+1)^2 - (k+1)$ is even. We have

$$\begin{aligned}(k+1)^2 - (k+1) &= (k^2 + 2k + 1) - (k+1) \\ &= k^2 + k \\ &= \underbrace{(k^2 - k)}_{\text{even by induct. hypoth.}} + \underbrace{(2k)}_{\text{even by def. even}} \qquad \text{(subtract, add } n\text{)}\end{aligned}$$

And the sum of two even numbers is even.

Hence, $k^2 - k$ is even $\implies (k+1)^2 - (k+1)$ is even.

## Exercise 2

Show by induction that $2^n < n!$ for all $n \geq 4$.

## Solution

**Induction base.** Take $n = 4$. Then $2^n = 2^4 = 16$ and $n! = 4! = 4 \cdot 3 \cdot 2 \cdot 1 = 24$. So $2^n < n!$ when $n = 4$.

**Induction step.** Now we assume the proposition holds at index $k$ (this is called the induction hypothesis), and we need to show it holds at index $k+1$. That is, we assume $2^k < k!$ (this is called the induction hypothesis), and we need to show $2^{k+1} < (k+1)!$ We have

$$\begin{aligned} 2^{k+1} &= 2 \cdot 2^k \\ &< 2 \cdot k! && \text{(by the induction hypothesis)} \\ &< (k+1) \cdot k! && \text{(since } k \geq 4\text{)} \\ &= (k+1)! && \text{(by def. factorial)} \end{aligned}$$

Hence, $2^k < k! \implies 2^{k+1} < (k+1)!$

## Exercise 3

Show by induction that $f_0 + \cdots + f_n = f_{n+2} - 1$ in the Fibonacci sequence.

## Solution

**Induction base.** Take $n = 0$. Then $f_0 + \cdots + f_n = f_0 = 0$ and $f_{n+2} - 1 = f_2 - 1 = 1 - 1 = 0$. So $f_0 + \cdots + f_n = f_{n+2} - 1$ when $n = 0$.

**Induction step.** Now we assume the proposition holds at index $k$ (this is called the induction hypothesis), and we need to show it holds at index $k + 1$. That is, we assume $f_0 + \cdots + f_k = f_{k+2} - 1$, and we need to show that $f_0 + \cdots + f_{k+1} = f_{k+3} - 1$. We have

$$\begin{aligned} f_0 + \cdots + f_{k+1} &= (f_0 + \cdots + f_k) + f_{k+1} \\ &= (f_{k+2} - 1) + f_{k+1} && \text{(by the induction hypothesis)} \\ &= (f_{k+1} + f_{k+2}) - 1 \\ &= f_{k+3} - 1 && \text{(by def. Fibonacci number)} \end{aligned}$$

Hence, if the proposition holds at index $k$, it holds at index $k + 1$.

## Exercise 4

Show by induction that $f_n < 2^n$ in the Fibonacci sequence.

## Solution

**Induction base.** Take $n = 0$. Then $f_n = f_0 = 0$ and $2^n = 2^0 = 1$. So $f_n < 2^n$ when $n = 0$.

**Induction step.** Here we use strong induction. We assume the proposition holds at indices $1, 2, \ldots k$ (this is the (strong) induction hypothesis), and we need to show it holds at index $k + 1$. That is, we assume $f_j < 2^j$ for $j = 1, \ldots, k$, and we need to show that $f_{k+1} < 2^{k+1}$. We have

$$
\begin{aligned}
f_{k+1} &= f_k + f_{k-1} && \text{(by def. Fibonacci number)} \\
&< 2^k + 2^{k-1} && \text{(by the (strong) induction hypothesis)} \\
&< 2^k + 2^k \\
&= 2 \cdot 2^k \\
&= 2^{k+1}.
\end{aligned}
$$

Hence, $f_k < 2^k \implies f_{k+1} < 2^{k+1}$.